








Advanced Time Series Analysis and Forecasting

Seminar 7: Regime-switching models



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **EM**: Expectation–Maximisation (algorithm)
- **EODHD**: EOD Historical Data (market-data provider)
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **GNP**: Gross National Product
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HICP**: Harmonised Index of Consumer Prices
- **HMP**: Haas–Mittnik–Paolella (MS-GARCH, 2004)
- **HQ**: Hannan–Quinn (information criterion)

Acronyms used in this material (2/2)

- **LR**: Likelihood Ratio (test)
- **MCMC**: Markov Chain Monte Carlo
- **ML**: Machine Learning
- **MS**: Markov Switching
- **MS-GARCH**: Markov-Switching GARCH
- **MS-VAR**: Markov-Switching Vector AutoRegression
- **MSIAH**: Markov-Switching Intercept, Autoregressive parameters and Heteroskedasticity
- **MSIAH-AR**: Markov-Switching Intercept, AR parameters and Heteroskedasticity AutoRegression
- **MSIH**: Markov-Switching Intercept and Heteroskedasticity
- **MSM**: Markov-Switching Mean (Krolzig notation)
- **MSM-AR**: Markov-Switching Mean AutoRegression
- **NBER**: National Bureau of Economic Research
- **OLS**: Ordinary Least Squares
- **QPS**: Quadratic Probability Score
- **TVTP**: Time-Varying Transition Probabilities
- **US**: United States

Today's question and seminar structure

- The question of the seminar
  - **how do we infer an unobserved regime, estimate a switching model, decide how many regimes there are, and read the result?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the Hamilton filter by hand; the Kim smoother and joint probabilities	Solved, Proposed	A1
A3, A4	one EM step; regime forecasts and persistence	Solved, Proposed	A1, A3
A5, A6	information criteria and a bootstrap p-value; identification and the expanded state	Solved, Proposed	A5
A7, A8	MS-GARCH variances (Haas–Mittnik–Paolella, Gray); Gibbs conditionals	Solved, Proposed	A7, A3
B1, B2	numpy against statsmodels (US GDP); Hamilton (1989)	Solved, Proposed	B1
B3, B4	how many regimes in Romanian GDP growth?; Romanian inflation regimes	Solved, Proposed	B3
B5, B6	bull and bear regimes of the S&P 500 and the BET; MS-GARCH for the BET	Solved, Proposed	B5
C1, C2	EUR/RON in 2025; what is wrong in an AI answer?	Proposed	B5, A1–A8

- **[Solved]**: the full solution is in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it following the model, and the solution is discussed in class
- Part A: calculations and derivations on paper; Part B: real data, solved in the notebook, each ending with an interpretation question; Part C: open questions, including the critique of an AI answer

Data used

Series	Source	Frequency	Period
US real GDP, NBER recession quarters	FRED (GDPC1, USRECQ)	quarterly	1947–2026
Hamilton's real GNP data	statsmodels	quarterly	1951–1984
Romanian real GDP; HICP	Eurostat	quarterly; monthly	1995–2026
S&P 500, BET	EODHD	daily, weekly	2000–2026
EUR/RON	BNR reference rate	daily	2023–2026

- In the notebook: `hamilton_filter`, `kim_smoother`, `em_msr`, `best_em`, `msr_ml`, `msm_fit`, `lr_bootstrap`, `msgarch_fit`, `gibbs_ms`; no account or key is needed

Prerequisites for Today (1/26): the switching regression

- **MS(K) regression**: the coefficients and the variance depend on a hidden regime

$$y_t = x_t'\beta_{S_t} + \sigma_{S_t}\varepsilon_t, \qquad \varepsilon_t \text{ i.i.d. } N(0,1)$$

  - $S_t \in \{1, \ldots, K\}$: the regime at t, not observed; K: the number of regimes
  - x_t: the regressors (constant, lags of y_t); β_j: the coefficients in regime j
  - σ_j: the s.d. of the shock in regime j; given $S_t = j$, $y_t \sim N(x_t'\beta_j, \sigma_j^2)$

- **Density of y_t in regime j**, the input of the filter

$$\eta_{jt} = f(y_t \mid S_t = j, Y_{t-1}) = \frac{1}{\sigma_j}\,\phi\left(\frac{y_t - x_t'\beta_j}{\sigma_j}\right), \qquad \phi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}$$

  - ϕ: the standard Normal density; $Y_{t-1} = (y_1, \ldots, y_{t-1})$: the data up to $t-1$
  - a large η_{jt}: the value y_t is typical of regime j

Prerequisites for Today (2/26): special forms

- **MSIH(K)-AR(p)** Krolzig (1997): the intercept and the variance switch, the AR coefficients do not

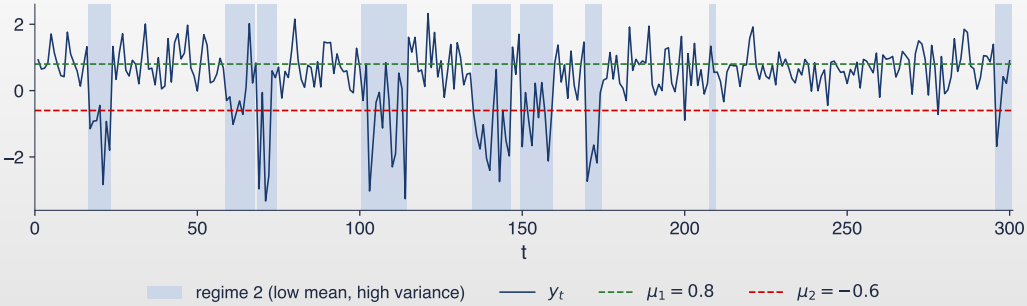
$$y_t = \nu_{S_t} + \sum_{k=1}^{p} \phi_k \, y_{t-k} + \sigma_{S_t} \varepsilon_t$$

  - ν_j: the intercept of regime j; ϕ_k: the AR coefficients, common to all regimes; p: the number of lags
- **MSIAH**: the AR coefficients switch too, ϕ_{k,S_t}
  - AR(1): level of regime $j = \nu_j/(1-\phi_j)$, the mean if the series stayed in regime j
  - ϕ_j close to 1: shocks persist for long inside regime j
- **MSM** (Hamilton 1989): the mean switches; the deviations from it follow an AR(p)

$$y_t - \mu_{S_t} = \sum_{k=1}^{p} \phi_k \, (y_{t-k} - \mu_{S_{t-k}}) + \sigma \varepsilon_t$$

  - μ_j: the mean of regime j; a switch moves the mean at once (in MSIH, gradually, through the AR terms)
  - η_t depends on $S_t, S_{t-1}, \ldots, S_{t-p}$: the filter must track this joint (expanded) state

Prerequisites for Today (3/26): a simulated two-regime model



- Simulated MSIH(2): $\mu = (0.8, -0.6)$, $\sigma = (0.6, 1.4)$, $p_{11} = 0.95$, $p_{22} = 0.85$, $T = 300$
- Shaded: the periods in regime 2, with a lower mean and a higher variance; turbulent periods come in spells
- In data only y_t is seen: the shading must be inferred, by the Hamilton filter below

Prerequisites for Today (4/26): the Markov chain

- **Transition probabilities** and the transition matrix

$$p_{ij} = \Pr(S_t = j \mid S_{t-1} = i), \qquad \mathrm{P} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}, \qquad \sum_j p_{ij} = 1$$

  - Markov property: the next regime depends only on the current one; p_{ii}: the probability of staying in i

- **Ergodic probabilities** $\boldsymbol{\pi}$: the long-run share of time in each regime, $\mathrm{P}'\boldsymbol{\pi} = \boldsymbol{\pi}$

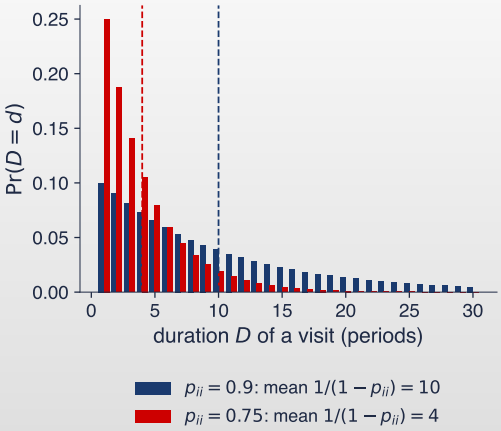
$$\pi_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}, \qquad \pi_2 = 1 - \pi_1$$

- **Duration** D_i of a visit to regime i: a geometric distribution

$$\Pr(D_i = d) = (1 - p_{ii})\, p_{ii}^{d-1}, \quad d = 1, 2, \ldots, \qquad \mathbb{E}[D_i] = \frac{1}{1 - p_{ii}}$$

  - $\mathbb{E}$: the expected value; $d - 1$ stays followed by one exit

Prerequisites for Today (5/26): the duration of a regime



- Bars: $\Pr(D = d)$ for $p_{ii} = 0.9$ and 0.75
- Dashed lines: the mean durations 10 and 4
- The most likely duration is always one period
- Near 1, small changes of p_{ii} change the duration a lot
  - $p_{ii} = 0.95$: 20 periods; $p_{ii} = 0.99$: 100 periods

Prerequisites for Today (6/26): regime forecasts

- *h*-**step forecast** of the regime probabilities

$$\hat{\xi}_{t+h|t} = (\mathrm{P}')^h \, \hat{\xi}_{t|t}$$

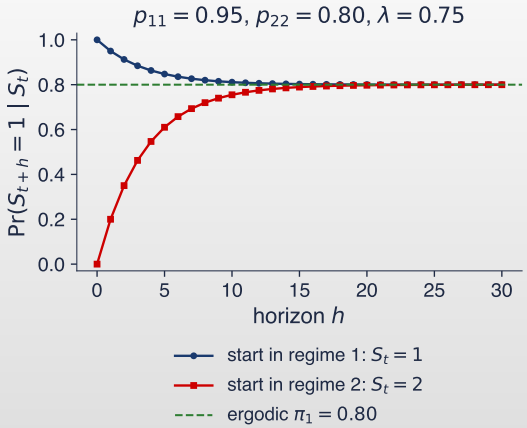
  - $\hat{\xi}_{t|t}$: the vector of $\Pr(S_t = j \mid Y_t)$, $j = 1, \ldots, K$; $(\mathrm{P}')^h$: the transitions over h periods
- **Two regimes**: persistence $\lambda = p_{11} + p_{22} - 1$

$$\Pr(S_{t+h} = 1 \mid S_t = 1) = \pi_1 + \lambda^h (1 - \pi_1), \qquad \Pr(S_{t+h} = 1 \mid S_t = 2) = \pi_1 - \lambda^h \pi_1$$

  - $|\lambda| < 1$: the forecasts converge to π_1; the gap shrinks by the factor λ each period
  - λ close to 1: long, sticky regimes and slow convergence
- **Forecast of the mean**: regime means weighted by the regime forecasts

$$\mathbb{E}[y_{t+h} \mid S_t = i] = \sum_{j=1}^{K} \Pr(S_{t+h} = j \mid S_t = i) \, \mu_j$$

Prerequisites for Today (7/26): convergence of regime forecasts



- $p_{11} = 0.95$, $p_{22} = 0.80$: $\lambda = 0.75$, $\pi_1 = 0.80$
- From $S_t = 1$ the probability falls, from $S_t = 2$ it rises; both reach π_1
- After h periods the gap is λ^h times the initial one: $0.75^{10} \approx 0.06$
- Beyond a few durations a regime forecast carries no information: only $\boldsymbol{\pi}$ is left

Prerequisites for Today (8/26): the Hamilton filter (1/2)

- **Notation**: $\hat{\xi}_{t|s}$ is the vector of $\Pr(S_t = j \mid Y_s)$, $j = 1, \ldots, K$
  - $s = t - 1$: predicted; $s = t$: filtered; $s = T$: smoothed (T: the sample length)
- **Step 1**: prediction with the chain

$$\hat{\xi}_{t|t-1} = \mathrm{P}' \, \hat{\xi}_{t-1|t-1}$$

- **Step 2**: update with Bayes' rule Hamilton (1989)

$$\hat{\xi}_{t|t} = \frac{\hat{\xi}_{t|t-1} \odot \eta_t}{f(y_t \mid Y_{t-1})}, \qquad f(y_t \mid Y_{t-1}) = \sum_{j=1}^{K} \hat{\xi}_{j,t|t-1} \, \eta_{jt}$$

  - $\eta_t = (\eta_{1t}, \ldots, \eta_{Kt})'$; $\odot$: the element-wise product; the denominator makes the probabilities sum to 1
  - posterior $\propto$ prior $\times$ likelihood: the regime in which y_t is typical gains probability

Prerequisites for Today (9/26): the Hamilton filter (2/2)

- **Log-likelihood**: the sum of the one-step predictive log densities

$$\ln L(\theta) = \sum_{t=1}^{T} \ln f(y_t \mid Y_{t-1})$$

  - θ: all parameters (β_j, σ_j, p_{ij}); maximised numerically or by EM

- **Start**: $\hat{\xi}_{1|0} = \boldsymbol{\pi}$, the ergodic probabilities

$$\hat{\xi}_{j,1|1} = \frac{\pi_j \eta_{j1}}{\pi_1 \eta_{11} + \pi_2 \eta_{21}}, \qquad j = 1, 2$$

  - the first observation moves the prior π_j towards the regime whose density at y_1 is larger

- **Filtered** probabilities use only $y_1, \ldots, y_t$
  - they show what a forecaster could know in real time (with the data available then)

Prerequisites for Today (10/26): the Kim smoother and the predictive density

- **Kim smoother** Kim (1994): runs backwards from $\hat{\xi}_{T|T}$

$$\Pr(S_t = i, S_{t+1} = j \mid Y_T) = \frac{\hat{\xi}_{i,t|t}\, p_{ij}\, \hat{\xi}_{j,t+1|T}}{\hat{\xi}_{j,t+1|t}}, \qquad \hat{\xi}_{i,t|T} = \sum_{j=1}^{K} \Pr(S_t = i, S_{t+1} = j \mid Y_T)$$

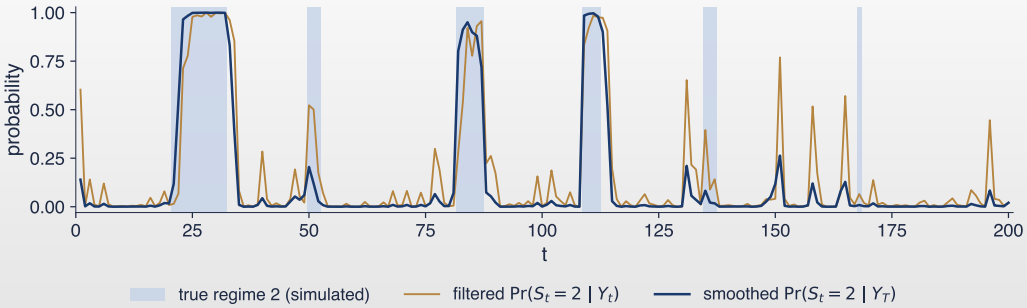
  - the joint probabilities of two consecutive regimes are the input of the EM step
  - smoothed = with hindsight; real-time dating is never evaluated with smoothed probabilities

- **Predictive density** one step ahead: a mixture of the regime densities

$$f(y_{T+1} \mid Y_T) = \sum_{j=1}^{K} \hat{\xi}_{j,T+1|T}\, \frac{1}{\sigma_j}\, \phi\left(\frac{y_{T+1} - x'_{T+1}\beta_j}{\sigma_j}\right), \qquad \hat{\xi}_{T+1|T} = \mathrm{P}'\hat{\xi}_{T|T}$$

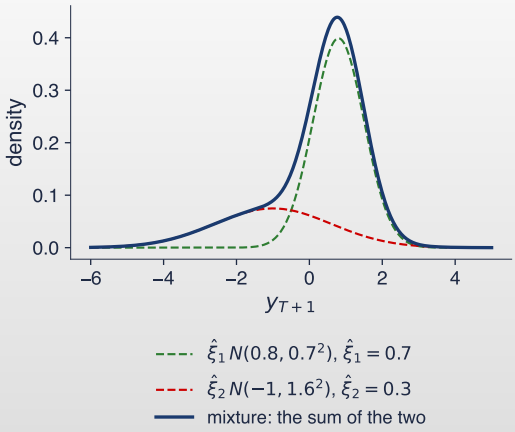
  - the weights are the predicted regime probabilities; the mixture can be skewed or bimodal

Prerequisites for Today (11/26): filtered and smoothed probabilities



- Simulated: $\mu = (1, -1)$, $\sigma_1 = \sigma_2 = 1$, $p_{11} = 0.95$, $p_{22} = 0.90$, $T = 200$; filter and smoother with the true parameters
- Filtered (amber): reacts with a delay and gives false alarms after single outliers
- Smoothed (blue): also uses later data, so it is sharper and closer to the true regime

Prerequisites for Today (12/26): the predictive mixture



- Weights 0.7 and 0.3; regime densities $N(0.8, 0.7^2)$ and $N(-1, 1.6^2)$
- Each dashed curve is already multiplied by its weight; the mixture is their sum
- The turbulent regime gives a long left tail: the mixture is skewed to the left
- A point forecast (the mean) hides this shape: report the density

Prerequisites for Today (13/26): regime dating and data transformations

- **QPS** (quadratic probability score) Diebold and Rudebusch (1989): distance between a regime probability and an event

$$\text{QPS} = \frac{2}{T}\sum_{t=1}^{T}(p_t - d_t)^2 \in [0, 2]$$

  - p_t: the probability of the low regime at t; $d_t = 1$ in an NBER recession quarter, 0 otherwise
  - 0: perfect dating; a constant $p_t = 0.5$ gives 0.5; 2: always wrong with certainty
- **Growth rates and returns**: $y_t = 100\,(\ln X_t - \ln X_{t-1})$, in percent
  - X_t: GDP, a price index or an index level; annual inflation: $100\,(\ln X_t - \ln X_{t-12})$ for monthly data
- **Annualisation** with A periods per year ($A = 52$ weekly, $A = 252$ daily)
  - mean: $A\,\mu_j$; volatility: $\sqrt{A}\,\sigma_j$ (the variance adds up over independent periods)

Prerequisites for Today (14/26): the EM algorithm

- **EM** Hamilton (1990), Dempster, Laird and Rubin (1977): E-step = the smoothed probabilities (Kim); M-step = weighted averages

$$\hat{p}_{ij} = \frac{\sum_{t=2}^{T} \Pr(S_{t-1} = i, S_t = j \mid Y_T)}{\sum_{t=2}^{T} \Pr(S_{t-1} = i \mid Y_T)}$$

  - the expected number of $i \to j$ transitions over the expected number of visits to i
- **Regime parameters** (constant only): weighted mean and variance, weights $\hat{\xi}_{j,t|T}$

$$\hat{\mu}_j = \frac{\sum_t \hat{\xi}_{j,t|T}\, y_t}{\sum_t \hat{\xi}_{j,t|T}}, \qquad \hat{\sigma}_j^2 = \frac{\sum_t \hat{\xi}_{j,t|T}\, (y_t - \hat{\mu}_j)^2}{\sum_t \hat{\xi}_{j,t|T}}$$

  - with regressors: weighted least squares for β_j, with the same weights
- **Convergence**: no step lowers $\ln L$, but EM stops at a local maximum
  - or at a degenerate one ($\sigma_j \to 0$ on a single point): use many starting values, then numerical ML

Prerequisites for Today (15/26): testing the number of regimes

- **LR statistic** (likelihood ratio) of $K = 1$ against $K = 2$

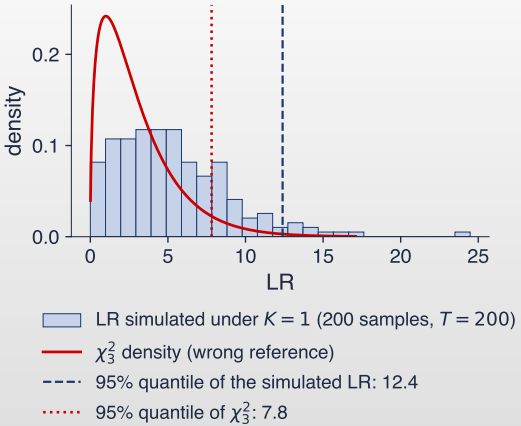
$$LR = 2\left[\ln L(\hat{\theta}_{K=2}) - \ln L(\hat{\theta}_{K=1})\right] \geq 0$$

  - $\hat{\theta}_K$: the ML estimates with K regimes; a large LR: the second regime improves the fit
- **The usual theory fails** Hansen (1992), Garcia (1998)
  - under H_0 ($K = 1$), p_{11} and p_{22} are not identified, the null is on the boundary, the score is zero
  - so LR does not follow χ^2_q (q: the number of extra parameters); the χ^2_q critical values are too small
- **Parametric bootstrap**: simulate B samples from the fitted $K = 1$ model, re-estimate both models

$$p = \frac{1 + \#\{b : LR^*_b \geq LR\}}{B + 1}$$

  - LR^*_b: the statistic in bootstrap sample b; $\#$: the number of samples; reject $K = 1$ at 5% if $p < 0.05$

Prerequisites for Today (16/26): the distribution of LR under one regime



- 200 samples of $T = 200$ from $N(0, 1)$: $K = 1$ is true
- Each sample: $K = 1$ and MSIH(2) estimated (EM from 4 starts), then LR
- 95% quantile: 12.4 simulated against 7.8 for χ^2_3
  - with χ^2_3 a true null is rejected far more often than 5%
- The bootstrap reproduces this distribution for the model and sample at hand

Prerequisites for Today (17/26): information criteria and identification

- **Information criteria**: fit minus a penalty for size; the smallest value wins

$$\text{AIC} = -2\ln L + 2k, \qquad \text{BIC} = -2\ln L + k\ln T, \qquad \text{HQ} = -2\ln L + 2k\ln\ln T$$

  - k: the number of free parameters, including the $K(K-1)$ free transition probabilities
  - penalty per parameter: 2 (AIC), $\ln T$ (BIC), $2\ln\ln T$ (HQ); BIC is the most parsimonious for $T \geq 8$
  - the criteria need no identification under H_0, but they are a guide, not a test with a known size
- **Label switching**: renaming the regimes leaves L unchanged
  - the likelihood has several equivalent maxima; name the regimes by an ordering ($\mu_1 < \mu_2$ or $\sigma_1 < \sigma_2$)

Prerequisites for Today (18/26): GARCH and the path dependence of MS-GARCH

- **GARCH(1,1)** Bollerslev (1986): the variance reacts to the last shock and to its own past

$$\varepsilon_t = \sqrt{h_t}\, z_t, \qquad h_t = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, h_{t-1}$$

  - z_t i.i.d. $N(0,1)$; h_t: the conditional variance; $\omega > 0$, $\alpha, \beta \geq 0$
  - α: reaction to news; β: memory; persistence $\alpha + \beta < 1$; long-run variance $\omega/(1-\alpha-\beta)$
  - half-life of a variance shock: $\ln 0.5 / \ln(\alpha + \beta)$ periods

- **Naive MS-GARCH**: all parameters switch with S_t

$$h_t = \omega_{S_t} + \alpha_{S_t} \varepsilon_{t-1}^2 + \beta_{S_t} h_{t-1}$$

  - ω_j, α_j, β_j: GARCH parameters of regime j; h_{t-1} depends on S_{t-1}, which depends on S_{t-2}, ...
  - h_t depends on the whole regime path: the likelihood would need one term per possible history

Prerequisites for Today (19/26): two tractable MS-GARCH models

- **The HMP model** Haas, Mittnik and Paolella (2004): K GARCH processes run in parallel

$$h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j\, h_{j,t-1}, \quad j = 1, \ldots, K, \qquad \varepsilon_t = \sqrt{h_{S_t,t}}\, z_t$$

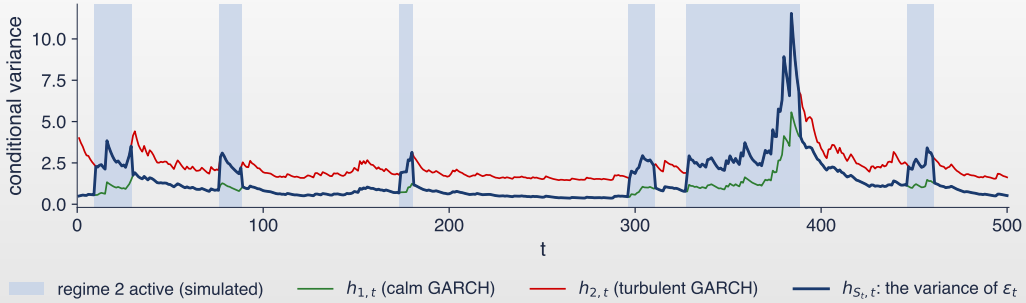
  - each $h_{j,t}$ uses its own lag: it does not depend on the regime path; the Hamilton filter applies
  - persistence of regime j: $\alpha_j + \beta_j$

- **Gray's model** Gray (1996): the history is collapsed into one number h_{t-1}

$$h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j\, h_{t-1}, \qquad h_{t-1} = \sum_{j=1}^{K} \Pr(S_{t-1} = j \mid Y_{t-2})\, h_{j,t-1}$$

  - h_{t-1}: the variance of the mixture given Y_{t-2} (with a mean common to the regimes)

Prerequisites for Today (20/26): MS-GARCH variances in parallel



- Simulated HMP model: $(\omega, \alpha, \beta) = (0.02, 0.04, 0.92)$ calm and $(0.20, 0.10, 0.85)$ turbulent; $p_{\mathbf{11}} = 0.99$, $p_{\mathbf{22}} = 0.97$
- Both variances are updated every period with the same shock; the variance of ε_t (blue) jumps to $h_{\mathbf{2},t}$ when regime 2 starts
- Persistence inside the regimes plus persistence of the regimes: a single GARCH fitted to such data has $\alpha + \beta$ near 1

Prerequisites for Today (21/26): Gibbs sampling: the transition probabilities

- **Gibbs sampler** Albert and Chib (1993), Chib (1996): draws each block from its full conditional
  - full conditional: the distribution of one block given all other blocks and the data
  - $S \mid \theta, Y$: forward filtering, then backward sampling of S_t with $\Pr(S_t = i \mid S_{t+1} = j, Y_t) \propto \hat{\xi}_{i,t|t}\, p_{ij}$
- **Beta prior** $p_{11} \sim \text{Beta}(a, b)$, density $\propto p^{a-1}(1-p)^{b-1}$, is conjugate

$$p_{11} \mid S \sim \text{Beta}(a + n_{11},\ b + n_{12})$$

  - n_{ij}: the number of $i \to j$ transitions in the drawn path; a, b: prior parameters, worth $a + b$ extra transitions
- **Moments of** $\text{Beta}(c, d)$

$$\mathbb{E} = \frac{c}{c+d}, \qquad \text{Var} = \frac{cd}{(c+d)^2(c+d+1)}$$

Prerequisites for Today (22/26): Gibbs sampling: the regime means

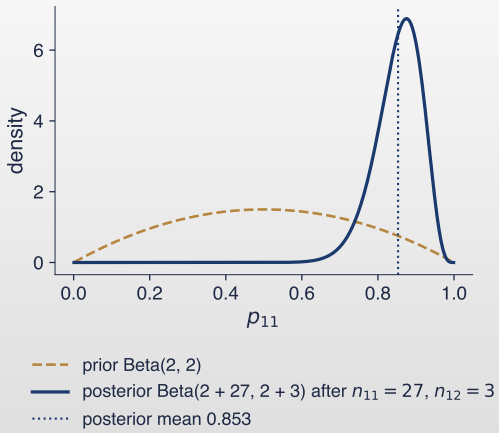
- **Normal prior** $\mu_j \sim N(m_0, v_0)$, σ_j^2 known: the posterior is Normal

$$\mu_j \mid S, Y \sim N(m_j, v_j), \qquad \frac{1}{v_j} = \frac{1}{v_0} + \frac{n_j}{\sigma_j^2}, \qquad m_j = v_j \left(\frac{m_0}{v_0} + \frac{n_j \bar{y}_j}{\sigma_j^2} \right)$$

  - m_0, v_0: prior mean and variance; n_j: the number of periods in regime j; $\bar{y}_j$: their average
  - precision = 1/variance: precisions add; m_j is a precision-weighted average of m_0 and $\bar{y}_j$
  - vague prior (v_0 large): $m_j \approx \bar{y}_j$ and $v_j \approx \sigma_j^2 / n_j$

- **Label switching in MCMC** Frühwirth-Schnatter (2001)
  - the sampler may swap the regimes between draws; identify them afterwards by an ordering

Prerequisites for Today (23/26): Bayesian updating of a transition probability



- Prior Beta(2, 2): symmetric around 0.5 and vague
- A drawn path with $n_{11} = 27$, $n_{12} = 3$: posterior Beta(29, 5), mean 0.853
- 30 transitions out of regime 1 already concentrate the posterior
- Any function of p_{11}, e.g. the duration $1/(1 - p_{11})$, gets a posterior by transforming the draws

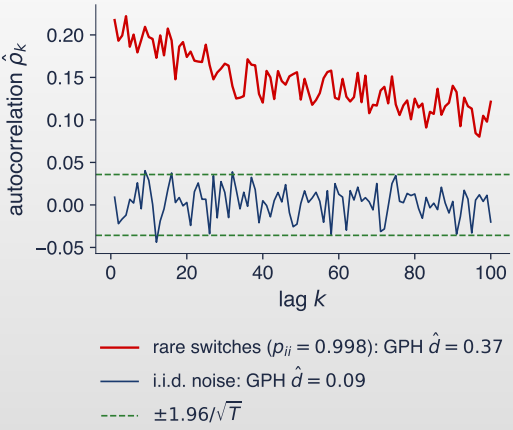
Prerequisites for Today (24/26): long memory and the GPH estimator

- **Long memory**: the autocorrelations decay hyperbolically, not geometrically
  - $\rho_k \approx c\, k^{2d-1}$ for large k; ρ_k: the autocorrelation at lag k; $0 < d < 1/2$: the memory parameter
- **GPH**: log-periodogram regression on the m lowest frequencies

$$\ln I(\lambda_j) = c - 2d\, \ln\big(2\sin(\lambda_j/2)\big) + u_j, \qquad \lambda_j = \frac{2\pi j}{T}, \quad j = 1, \ldots, m$$

  - $I(\lambda) = \left|\sum_t y_t e^{-i\lambda t}\right|^2/(2\pi T)$: the periodogram; $\hat{d}$ = OLS slope divided by -2; usually $m = \sqrt{T}$
  - $\hat{d} \approx 0$: short memory; $\hat{d} > 0$: a slowly decaying ACF, i.e. much power at low frequencies
- **Regime switching** Diebold and Inoue (2001): rare switches also put power at low frequencies
  - a positive $\hat{d}$ alone cannot tell long memory from regimes (chart on the next slide)

Prerequisites for Today (25/26): rare regime switches and spurious long memory



- Red: mean ± 0.5 switching with $p_{11} = p_{22} = 0.998$, plus $N(0,1)$ noise, $T = 3\,000$
- Blue: i.i.d. $N(0,1)$ noise of the same length
- The switching series has autocorrelations far above the band even at lag 100
- Its GPH estimate is clearly positive, although no long-memory process generated it

Prerequisites for Today (26/26): time-varying transitions and breaks

- **TVTP** (time-varying transition probabilities) Filardo (1994): p_{ii} depends on an observed driver

$$p_{ii,t} = \frac{1}{1 + \exp\big(-(\gamma_{0i} + \gamma_{1i}\, z_{t-1})\big)}$$

  - z_{t-1}: a leading indicator (e.g. news, an interest-rate spread); $\gamma_{1i} > 0$: a high z keeps regime i longer
- **Level break** at an unknown date τ: a one-off shift that does not recur

$$y_t = \mu + \delta\, 1\{t \geq \tau\} + \varepsilon_t$$

  - $1\{\cdot\}$: 1 if the condition holds, 0 otherwise; δ: the size of the shift
  - a Markov regime recurs with probability $1 - p_{ii} > 0$ each period; a break is a regime with $p_{22} = 1$

A1: the chain and the first filter step [Solved]

Task

- $y_t = \mu_{S_t} + \varepsilon_t$, $\varepsilon_t \sim N(0,1)$, $\mu_{\mathbf{1}} = 1$ (expansion), $\mu_{\mathbf{2}} = -1$ (recession), $p_{\mathbf{11}} = 0.9$, $p_{\mathbf{22}} = 0.75$; $y_{\mathbf{1}} = 0.5$.
- 1. Compute the ergodic probabilities and the expected durations.
- 2. With $\hat{\xi}_{\mathbf{1}|\mathbf{0}} = \boldsymbol{\pi}$, compute $\eta_{\mathbf{1}}$, $f(y_{\mathbf{1}})$ and $\hat{\xi}_{\mathbf{1}|\mathbf{1}}$.
- 3. Give the first log-likelihood contribution.
- Report: two probabilities and two durations; two densities, one density and two probabilities; one number.

Solution

- 1. $\pi_{\mathbf{1}} = 0.25/0.35 = 0.714$, $\pi_{\mathbf{2}} = 0.286$; durations $1/0.1 = 10$ and $1/0.25 = 4$
- 2. $\eta_{\mathbf{1}} = (\phi(-0.5), \phi(1.5)) = (0.3521, 0.1295)$; $f(y_{\mathbf{1}}) = 0.714 \cdot 0.3521 + 0.286 \cdot 0.1295 = 0.2885$; $\hat{\xi}_{\mathbf{1}|\mathbf{1}} = (0.872, 0.128)$
- 3. $\ln f(y_{\mathbf{1}}) = -1.243$
- Interpretation: a growth of 0.5 is closer to the expansion mean, so the expansion probability rises from the prior 0.714 to 0.872

Back Exercise list

A2: the second step and the Kim smoother [Proposed]

Solution discussed in class

Task

- The model of A1 with $y_2 = -1.5$. Model: A1.
- 1. Compute $\hat{\xi}_{2|1} = \mathrm{P}'\hat{\xi}_{1|1}$ and $\hat{\xi}_{2|2}$.
- 2. Take $T = 2$ and compute the smoothed $\hat{\xi}_{1|2}$ with the Kim formula.
- 3. Compute the four joint probabilities $\Pr(S_1 = i, S_2 = j \mid Y_2)$ and check that they sum to 1.
- Report: two vectors, one vector, a 2×2 table and one sentence on why the smoothed expansion probability of period 1 fell.

A3: one EM step [Solved]

Task

- $T = 5$, $y = (1.2, 0.8, -0.9, -1.4, 0.6)$; smoothed $\hat{\xi}_{\mathbf{1},t|T} = (0.9, 0.8, 0.1, 0.05, 0.7)$; joint $\Pr(S_{t-\mathbf{1}} = 1, S_t = 1 \mid Y_T) = (0.75, 0.08, 0.04, 0.05)$ for $t = 2, \ldots, 5$.
- 1. Update $p_{\mathbf{11}}$.
- 2. Update $\mu_{\mathbf{1}}$, $\mu_{\mathbf{2}}$ (MSIH without regressors).
- 3. Update $\sigma^2_{\mathbf{1}}$, $\sigma^2_{\mathbf{2}}$.
- Report: one number, two numbers, two numbers.

Solution

- 1. $p_{\mathbf{11}} = 0.920/1.850 = 0.497$ (expected $1 \to 1$ transitions over expected visits to 1 at $t = 1..4$)
- 2. $\mu_{\mathbf{1}} = \sum \hat{\xi}_{\mathbf{1}t} y_t / \sum \hat{\xi}_{\mathbf{1}t} = 0.776$, $\mu_{\mathbf{2}} = -0.686$ (weights $1 - \hat{\xi}_{\mathbf{1}t}$)
- 3. $\sigma^2_{\mathbf{1}} = 0.275$, $\sigma^2_{\mathbf{2}} = 0.742$ (weighted mean squared deviations)
- Interpretation: each M-step is a set of weighted averages; with five points the estimate of $p_{\mathbf{11}}$ is very imprecise

Back Exercise list

A4: regime forecasts and persistence [Proposed]

Solution discussed in class

Task

- The chain and means of A1; today the economy is in recession ($S_t = 2$ known). Model: A1, A3.
- 1. Show that $\xi_t = 1\{S_t = 1\}$ follows $\xi_t = (1 - p_{22}) + \lambda \xi_{t-1} + v_t$ with $\mathbb{E}[v_t \mid S_{t-1}] = 0$; give λ.
- 2. Compute $\Pr(S_{t+h} = 1 \mid S_t = 2)$ and $\mathbb{E}[y_{t+h} \mid S_t = 2]$ for $h = 1, 4, 12$.
- 3. Give the limit as $h \to \infty$.
- Report: one derivation and one number; six numbers; one number.

Back Exercise list

A5: how many regimes? [Solved]

Task

- $T = 200$; $\ln L$ for $K = 1, 2, 3$: -310.4, -296.1, -292.8 with 3, 7, 13 parameters; a bootstrap of the LR for $K = 2$ against $K = 3$ gives 6 of 199 simulated values above the observed one.
- 1. Compute AIC, BIC and HQ for each K and the choice of each criterion.
- 2. Compute LR for 1 against 2 and for 2 against 3; say why χ^2_4 is not the right reference for the first.
- 3. Compute the bootstrap p-value.
- Report: a table of nine values and three choices; two numbers and one sentence; one number.

Solution

- 1. AIC: 626.8, 606.2, 611.6; BIC: 636.7, 629.3, 654.5; HQ: 630.8, 615.5, 629.0: all choose $K = 2$
- 2. LR $=$ 28.6 and 6.6; under $K = 1$ the transition probabilities are not identified and the score is zero, so the 9.49 of χ^2_4 is not valid
- 3. $p = (1 + 6)/(199 + 1) = 0.035$: the bootstrap rejects 2 regimes at 5% although all criteria prefer 2

Back Exercise list

A6: identification and the expanded state [Proposed]

Solution discussed in class

Task

- Hamilton's MSM(K)-AR(4) and the MSIAH(K)-AR(4), $K = 2$ and $K = 3$. Model: A5.
- 1. How many equivalent likelihood modes does label switching create?
- 2. How many states does the filter of MSM-AR(4) track, and why does MSIAH-AR(4) need only K?
- 3. Count the free parameters of both models.
- Report: two numbers; two numbers and one sentence; four numbers.

A7: MS-GARCH, one step by hand [Solved]

Task

- Two regimes, $(\omega, \alpha, \beta) = (0.02, 0.05, 0.9)$ and $(0.2, 0.1, 0.8)$; $\varepsilon_{t-\mathbf{1}} = 1.5$; $h_{\mathbf{1},t-\mathbf{1}} = 0.5$, $h_{\mathbf{2},t-\mathbf{1}} = 2$; $\Pr(S_{t-\mathbf{1}} = j \mid Y_{t-\mathbf{2}}) = (0.7, 0.3)$.
- 1. Compute $h_{\mathbf{1},t}$, $h_{\mathbf{2},t}$ in the Haas–Mittnik–Paolella model.
- 2. Compute the collapsed $h_{t-\mathbf{1}}$ and $h_{\mathbf{1},t}$, $h_{\mathbf{2},t}$ in Gray's model.
- 3. How many regime histories would the naive model need after 10 and after 100 periods?
- Report: two numbers; three numbers; two numbers.

Solution

- 1. $h_{\mathbf{1},t} = 0.02 + 0.05 \cdot 2.25 + 0.9 \cdot 0.5 = 0.5825$; $h_{\mathbf{2},t} = 0.2 + 0.1 \cdot 2.25 + 0.8 \cdot 2 = 2.0250$
- 2. $h_{t-\mathbf{1}} = 0.7 \cdot 0.5 + 0.3 \cdot 2 = 0.95$; $h_{\mathbf{1},t} = 0.9875$, $h_{\mathbf{2},t} = 1.1850$
- 3. $2^{\mathbf{10}} = 1024$, $2^{\mathbf{100}} \approx 1.27 \times 10^{\mathbf{30}}$
- Interpretation: Gray pulls both regime variances towards the mixture: the calm regime inherits turbulence and the turbulent one forgets it

Back Exercise list

A8: Gibbs full conditionals [Proposed]

Solution discussed in class

Task

- A draw of S has $n_{11} = 46$, $n_{12} = 4$ transitions out of regime 1, and $n_1 = 30$ observations in regime 1 with mean $\bar{y}_1 = 0.9$; prior $p_{11} \sim \text{Beta}(8, 2)$, $\mu_1 \sim N(0, 10)$, $\sigma_1^2 = 0.25$ fixed. Model: A3.
- 1. Give the full conditional of p_{11}, its mean, standard deviation and the implied duration.
- 2. Give the full conditional of μ_1.
- 3. Explain why the prior on p_{11} also protects against the degenerate maxima of ML.
- Report: a distribution and three numbers; a distribution; one sentence.

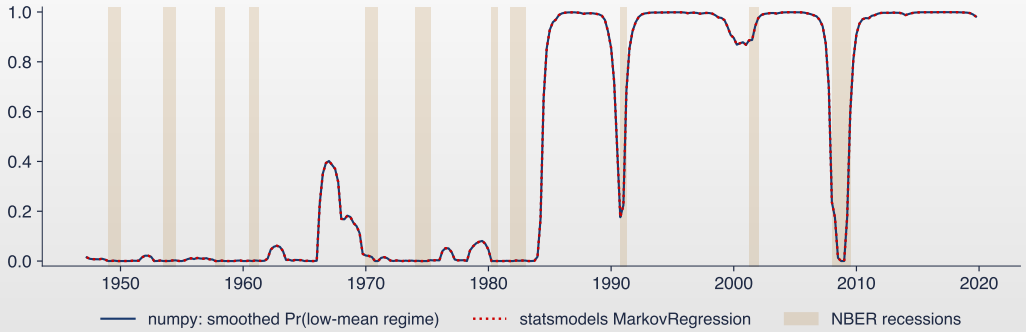
Back Exercise list

B1: numpy against statsmodels [Solved]

- **Question**: do a transparent numpy implementation and statsmodels give the same answer for US GDP growth?
- **Inputs**: 100 $\times$ log change of GDPC1, 1947Q2–2019Q4 ($T = 291$); two regimes in mean and variance, no AR term
- **Tasks**
  1. Estimate the model by EM from 15 random starts, then polish by numerical ML.
  2. Estimate it with statsmodels MarkovRegression (20 random searches) and compare parameters and log-likelihoods.
  3. Compare the smoothed probabilities and compute the QPS against the NBER quarters.
  4. Interpretation: are the regimes recessions?
- **Report**: two parameter sets, three log-likelihoods, one difference, one QPS, one sentence
- Notebook: section B1

Back Exercise list

B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

Estimate	μ_1	μ_2	p_{11}	p_{22}	$\ln L$
numpy (EM + ML)	0.743	0.808	0.975	0.978	−348.50
statsmodels	0.743	0.808	0.975	0.978	−348.50

- EM alone: $\ln L = -348.57$ after 31 iterations; largest difference of the smoothed probabilities $< 10^{-4}$; $\sigma = 0.440$ and 1.196; QPS of the low-mean regime 1.096
- Interpretation: the two implementations agree
  - the regimes have almost the same mean and last 39.3 and 44.7 quarters
  - they are volatility eras (before and after the mid-1980s), not recessions; the QPS against NBER is poor

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B2: Hamilton (1989) on his data and after 1984 [Proposed]

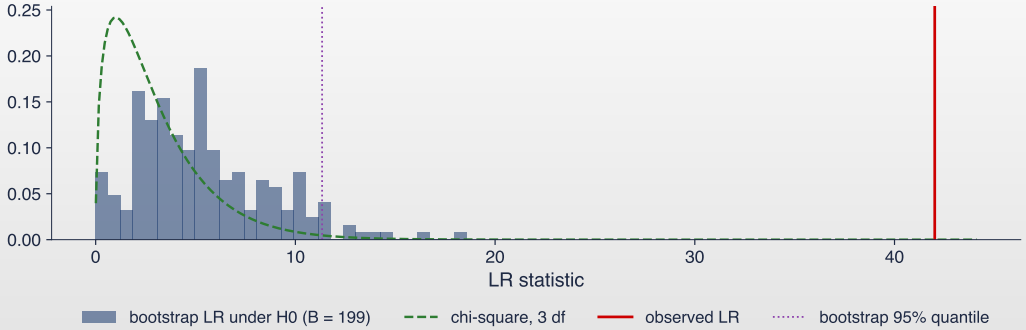
- **Question**: does Hamilton's model still find recessions in the Great Moderation?
- **Inputs**: Hamilton's real GNP data 1951Q2–1984Q4 (statsmodels); GDPC1 1984Q1–2019Q4; MSM(2)-AR(4); model: B1
- **Tasks**
  1. Estimate the model on Hamilton's data with the numpy expanded-state filter and with statsmodels MarkovAutoregression.
  2. Report the two regime means, the two persistence probabilities and the QPS against the NBER quarters.
  3. Re-estimate on 1984–2019 and report the same quantities and the share of quarters in the low regime.
  4. Interpretation: what does the low regime describe after 1984?
- **Report**: two log-likelihoods, two sets of five numbers, one share, one sentence
- Notebook: section B2

Back Exercise list

B3: how many regimes in Romanian GDP growth? [Solved]

- **Question**: is there statistical evidence for regimes in Romanian quarterly GDP growth?
- **Inputs**: Eurostat real GDP, seasonally and calendar adjusted, 100 $\times$ log change, 1995Q2–2026Q2 ($T = 125$); regimes in mean and variance
- **Tasks**
  1. Estimate $K = 1, 2, 3$ and compute AIC, BIC and HQ.
  2. Compute the LR statistic of $K = 1$ against $K = 2$ and its parametric bootstrap p-value ($B = 199$).
  3. Compare the bootstrap 95% quantile with the χ^2_3 critical value.
  4. Interpretation: which quarters form the low regime, and what does that say about the regimes?
- **Report**: a table of nine criteria, one LR and one p-value, two quantiles, a list of quarters, one sentence
- Notebook: section B3

B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

K	AIC	BIC	HQ
1	611.0	616.7	613.3
2	577.0	594.0	583.9
3	575.9	609.8	589.7

- LR = 42.01, bootstrap $p = 0.005$; bootstrap 95% quantile 11.34 against 7.81 for χ^2_3; regimes: $\mu = 0.03$ and 1.04, $\sigma = 3.92$ and 1.31, $p_{ii} = 0.90$ and 0.94
- Interpretation: the low regime contains 49 quarters
  - 1995Q2, 1995Q3, 1995Q4, 1996Q1, 1996Q2, 1996Q3, 1996Q4, 1997Q1, ...
  - the two regimes separate a volatile era with deep falls from stable growth, not recurring recessions

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B4: Romanian inflation regimes [Proposed]

- **Question**: do the inflation regimes differ only in level and volatility, or also in persistence?
- **Inputs**: Eurostat HICP, annual inflation $100 \ln(P_t / P_{t-\mathbf{12}})$, monthly from January 1997; three regimes; model: B3
- **Tasks**
  1. Estimate MSIH(3)-AR(1) (common AR coefficient) and MSIAH(3)-AR(1) (switching AR coefficient) by EM from 20 starts.
  2. Report the log-likelihoods, the BIC, the regime levels $\nu_j/(1-\phi_j)$, the AR coefficients and the durations.
  3. Report the filtered regime probabilities at the end of the sample.
  4. Interpretation: which model would you use to forecast inflation in 2027, and why?
- **Report**: two log-likelihoods and BICs, two sets of levels, coefficients and durations, two probability vectors, one sentence
- Notebook: section B4

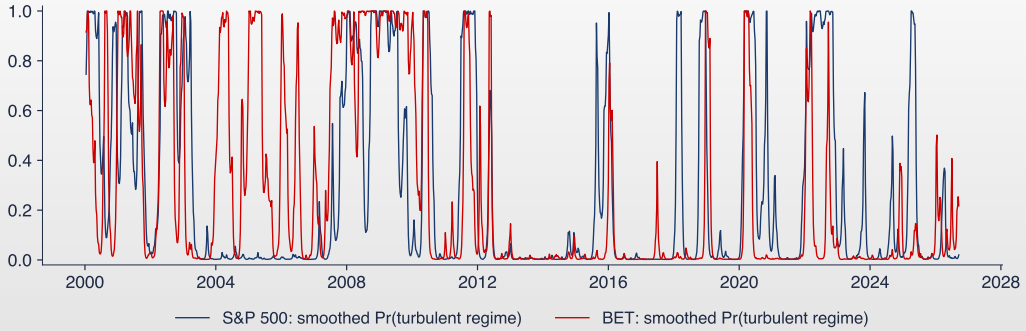
Back Exercise list

B5: bull and bear regimes of the S&P 500 and the BET [Solved]

- **Question**: are the turbulent regimes of the US and Romanian equity markets synchronised?
- **Inputs**: weekly log returns (Friday closes), 2000–2026, EODHD; two regimes in mean and variance, ordered by variance; model: B1
- **Tasks**
  1. Estimate the model for each index (EM from 15 starts and a numerical polish).
  2. Report annualised means and volatilities, expected durations and the share of turbulent weeks.
  3. Compute the correlation of the two turbulent probabilities and the maximum probability in 2008, 2020 and 2022.
  4. Interpretation: are BET regimes imported from the US?
- **Report**: a table of two rows and seven columns, one correlation, six probabilities, one sentence
- Notebook: section B5

Back Exercise list

B5: solution (1/2) [Solved]



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B5: solution (2/2) [Solved]

Index	μ calm	σ calm	μ turb.	σ turb.	durations (weeks)	% turb.	2008, 2020, 2022
S&P 500	16.6	10.9	-20.6	28.3	37; 14	26	1.00; 1.00; 1.00
BET	20.1	13.6	3.2	39.9	33; 12	25	1.00; 1.00; 1.00

- Correlation of the turbulent probabilities 0.48; both turbulent in 15% of weeks, BET alone in 11%
- Interpretation: the big global episodes are shared, but a large part of BET turbulence is domestic; regimes are mostly volatility regimes with imprecise means

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B6: MS-GARCH for the BET [Proposed]

- **Question**: how much of the GARCH persistence of the BET is regime persistence?
- **Inputs**: BET daily log returns (%), 2010–2026, $T = 4\,186$; GARCH(1,1) and the two-regime GARCH of Haas, Mittnik and Paolella; model: B5, A7
- **Tasks**
  1. Estimate both models by numerical ML with Normal innovations.
  2. Report $\alpha + \beta$ for GARCH and in each regime, the persistence probabilities, the regime volatilities and the BIC.
  3. Report the share of days in the high-volatility regime.
  4. Interpretation: what does this imply for a 10-day volatility forecast after a calm month?
- **Report**: two BICs, three persistences, two probabilities, two volatilities, one share, one sentence
- Notebook: section B6

Back Exercise list

C1: EUR/RON in 2025, a new regime or a break? [Proposed]

- **Question**: was the depreciation of May 2025 the start of a new volatility regime, a one-off level shift, or both?
- **Inputs**: BNR reference rate, daily log changes (%), January 2023 – September 2026 ($T = 926$); model: B5
- **Tasks**
  1. Write a five-line pre-registration: hypotheses, models (MSIH(2), a level break), sample, criteria, what would change your mind.
  2. Estimate the two-regime model and report the regime volatilities, durations, the share of turbulent days before and after May 2025.
  3. Interpretation: what can and cannot be concluded from one episode?
- **Report**: a five-line plan, a table of six numbers, a project plan
- Notebook: section C1

Back Exercise list

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise Markov-switching models. The answer:
- (a) The smoothed probabilities show what a forecaster knew in real time.
- (b) With p11 = 0.9 and p22 = 0.75, the ergodic probability of regime 1 is 0.9.
- (c) To test one regime against two, compare the LR statistic with a chi-square with 4 degrees of freedom.
- (d) EM always reaches the global maximum because each step increases the likelihood.
- (e) Hamilton's MSM-AR(4) with two regimes needs a filter over 32 states.
- (f) Gray's MS-GARCH is exact; the model of Haas, Mittnik and Paolella approximates the path.
- (g) A positive GPH estimate of d proves long memory rather than regime switching.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- The Hamilton filter is Bayes' rule applied each period; the Kim smoother runs it backwards
- EM is a sequence of weighted averages; start it many times
- The number of regimes is not tested with a χ^2 count
- Regimes found by a model are statistical objects: read them against events before naming them
- An AI answer is a draft: check filtered against smoothed, ergodic against persistence, local against global

After the seminar

- Lecture 7 develops each topic of today: derivations, testing, TVTP, MS-VAR, MS-GARCH, Bayesian estimation, long memory, forecasting
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: exchange-rate regimes of the leu, 1999–2026
- Reading: Hamilton (1989); Hamilton (1990); Kim (1994); Haas, Mittnik and Paolella (2004)
- **The seminar is for practice and is not graded**

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